

1. Create your AWS organization and add two AWS accounts, one with predefined AWS IAM Role and another with RoleForSwitchAccount named role.

Ans:

AWS accounts

Add an AWS account

The accounts listed below are members of your organization. The organization's management account is responsible for paying the bills for all accounts in the organization. You can use the tools provided by AWS Organizations to centrally manage these accounts. [Learn more](#)

3 requests to create AWS accounts have failed in the last 90 days.

Centralize root access for member accounts

You can delete root credentials for your member accounts and perform privileged actions from the management or delegated account. [Learn more about centralizing root access](#)

Enable in IAM

Organization

Actions

Organizational units (OUs) enable you to group several accounts together and administer them as a single unit instead of one at a time.

Search by name, email, account ID or OU ID.

HierarchyList

Organizational structure	Account created/joined date
Root r-6yon	
Piyush 097021215071 psblack+1@gmail.com	Created 2026/02/26
SiriusDEVmanagement account 596059883112 siriusdev19@gmail.com	Joined 2026/02/25

AWS accounts

[Add an AWS account](#)

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[Enable in IAM](#)

Organization

[Actions](#)

Organizational units (OUs) enable you to group several accounts together and administer them as a single unit instead of one at a time.

Search by name, email, account ID or OU ID.

[Hierarchy](#)[List](#)

Organizational structure

Account created/joined date

Root

r-6yon



Piyush

097021215071 | psblack+1@gmail.com

Created 2026/02/26



Piyush-DEV

895112955134 | siriusdev19+@gmail.com

Created 2026/03/01



Piyush-Prod

154335822557 | siriusdev19+1@gmail.com

Created 2026/03/01



SiriusDEV

management account

596059883112 | siriusdev19@gmail.com

Joined 2026/02/25

2. Illustrate how the switch role works.

Ans:

Switch Role

Switching roles enables you to manage resources across Amazon Web Services accounts using a single user. When you switch roles, you temporarily take on the permissions assigned to the new role. When you exit the role, you give up those permissions and get your original permissions back. [Learn more](#)

Account ID

The 12-digit account number or the alias of the account in which the role exists.

154335822557

IAM role name

The name of the role that you want to assume which can be found at the end of the role's ARN. For example, provide the **TestRole** role name from the following role ARN: `arn:aws:iam::123456789012:role/TestRole`.

OrganisationAccountAccessRole

Display name - optional

This name will appear in the console navigation bar when active. Choose a name to help identify the permission set assigned to the role.

OrganisationAccountAccessRole @ 154335822557

Display color - optional

The selected color displays in the console navigation when this role is active

None

[Cancel](#)[Switch Role](#)

Dashboard

Organization details

Organization ID
o-jkuqsduo

Management account email address
siriusdev19@gmail.com

Management account ID
596059883112

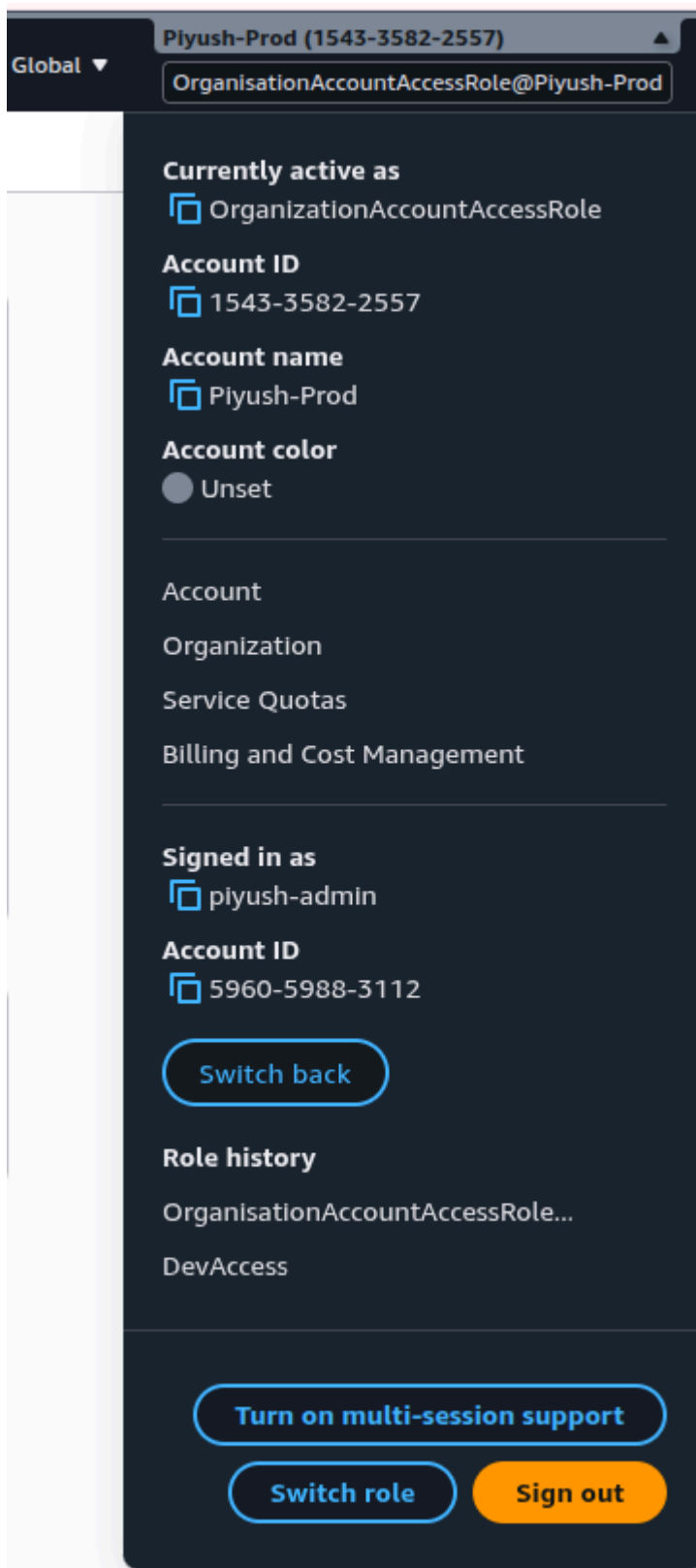
Feature set

Your organization has all features enabled. You can apply policies that can configure and limit what the accounts in the organization can do. Trusted AWS services can access your organization and accounts. The management account can create, manage and pay for the organization's accounts through consolidated billing.

Leave organization

[Leave this organization](#)

If you leave the organization, you become responsible for all billing charges related to this account. If you want to rejoin the organization, you must receive and approve a new invitation. [Learn more](#) 



3. Create a role in any of the AWS account that you have created in 2 question, so that you can switch from parent account to that child account; the role created must have permissions to access only s3 bucket.

Ans:

Select trusted entity Info

Trusted entity type

☐ AWS service

Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☒ AWS account

Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ Web identity

Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ SAML 2.0 federation

Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ Custom trust policy

Create a custom trust policy to enable others to perform actions in this account.

An AWS account

Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ This account (154335822557)

☒ Another AWS account

Account ID

Identifier of the account that can use this role

596059883112

Account ID is a 12-digit number.

Options

☐ Require external ID (Best practice when a third party will assume this role)

☐ Require MFA

Requires that the assuming entity use multi-factor authentication.

S3FullAccessForAccountSwitchRole Info

Delete

Summary

Creation date

March 01, 2026, 14:15 (UTC+05:30)

ARN

arn:aws:iam::154335822557:role/S3FullAccessForAccountSwitchRole

Link to switch roles in console

<https://signin.aws.amazon.com/switchrole?roleName=S3FullAccessForAccountSwitchRole&account=154335822557>

Last activity

-

Maximum session duration

1 hour

Edit

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (1) Info

You can attach up to 10 managed policies.



Simulate

Remove

Add permissions

Search

Filter by Type

All types

☐

Policy name

▲

Type

▼

Attached entities

▼

☐

AmazonS3FullAccess

AWS managed

1

Permissions boundary (not set)

Generate policy based on CloudTrail events

You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

Generate policy

No requests to generate a policy in the past 7 days.

S3FullAccessForAccountSwitchRole Info

Delete

Summary

Creation date

March 01, 2026, 14:15 (UTC+05:30)

ARN

arn:aws:iam::154335822557:role/S3FullAccessForAccountSwitchRole

Link to switch roles in console

<https://signin.aws.amazon.com/switchrole?roleName=S3FullAccessForAccountSwitchRole&account=154335822557>

Last activity

-

Maximum session duration

1 hour

Edit

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Trusted entities

Entities that can assume this role under specified conditions.

Edit trust policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Principal": {
7         "AWS": "arn:aws:iam::596059883112:root"
8       },
9       "Action": "sts:AssumeRole",
10      "Condition": {}
11    }
12  ]
13 }
```

Switch Role

Switching roles enables you to manage resources across Amazon Web Services accounts using a single user. When you switch roles, you temporarily take on the permissions assigned to the new role. When you exit the role, you give up those permissions and get your original permissions back. [Learn more](#)

Account ID

The 12-digit account number or the alias of the account in which the role exists.

IAM role name

The name of the role that you want to assume which can be found at the end of the role's ARN. For example, provide the **TestRole** role name from the following role ARN: `arn:aws:iam::123456789012:role/TestRole`.

Display name - optional

This name will appear in the console navigation bar when active. Choose a name to help identify the permission set assigned to the role.

Display color - optional

The selected color displays in the console navigation when this role is active

☐ NoneCancelSwitch Role

Search

[Alt+S]

United States (N. Virginia)

Piyush-Prod (1543-3582-2557)

S3FullAccessForAccountSwitchRole@Piyush-Prod

Buckets

Successfully created bucket "testbucketq2piysh"

View details

Buckets

General purpose buckets

All AWS Regions

Directory buckets

General purpose buckets (1)

Info

Copy ARN

Empty

Delete

Create bucket

Find buckets by name

< 1 >

Name	AWS Region	Creation date
testbucketq2piysh	US East (N. Virginia) us-east-1	March 1, 2026, 14:21:01 (UTC+05:30)

Account snapshot

Info

View dashboard

Updated daily

Storage Lens provides visibility into storage usage and activity trends.

External access summary

Info

Updated daily

External access findings help you identify bucket permissions that allow public access or access from other AWS accounts

4. Create a SCP that will restrict users not to create any resource in N.Virginia region.

Ans:

[Alt+S]

Global

SiriusDEV (5960-5988-3112)

piyush-admin

Service control policies have been enabled.

Disable service control policies

Service control policies

Info

Service control policies (SCPs) offer central control over the maximum available permissions for IAM users and IAM roles in an organization. [Learn more](#)

Available policies

Actions

Create policy

☐	Name	Kind	Description
☐	FullAWSAccess	AWS managed policy	Allows access to every operation

DenyRegion

[Delete](#)[Edit policy](#)

Policy details

Name

DenyRegion

ARN

arn:aws:organizations::596059883112:policy/o-jkuqsqduo/service_control_policy/p-74js3pzv

Policy type

Service control policy (customer managed)

Description

-

[Content](#)[Targets](#)[Tags](#)

Content

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "DenyNVirginiaRegion",
      "Effect": "Deny",
      "Action": "*",
      "Resource": "*",
      "Condition": {
        "StringEquals": {
          "aws:RequestedRegion": "us-east-1"
        }
      }
    }
  ]
}
```

5. Create 2 OUs with names as prod and uat; and move above created accounts in any of the OUs.

Ans:

prod

DeleteRename

An organization unit (OU) is a container for accounts within a root. An OU can contain other OUs up to five levels deep. This enables you to create an inverted tree hierarchy. The structure has a root at the top and branches of OUs that reach down. The branches end in accounts that act as the leaves of the tree. [Learn more](#)

Organizational unit details

Name

prod

ID

ou-6yon-dyayeo16

ARN

arn:aws:organizations::596059883112:ou/o-jkuqsqdueo/ou-6yon-dyayeo16

Children

Tags

Policies

Children

Actions

These are the child OUs and AWS accounts contained by this OU.

Organizational structure	Account created/joined date
This resource is empty	

✔ Successfully moved the AWS account 'Piyush-Prod' to the organizational unit 'prod'.✕

prodDeleteRename

An organization unit (OU) is a container for accounts within a root. An OU can contain other OUs up to five levels deep. This enables you to create an inverted tree hierarchy. The structure has a root at the top and branches of OUs that reach down. The branches end in accounts that act as the leaves of the tree. [Learn more](#)

Organizational unit details

Name

prod

ID

ou-6yon-dyayeo16

ARN

arn:aws:organizations::596059883112:ou/o-jkuqsqdueo/ou-6yon-dyayeo16

- Children
- Tags
- Policies

ChildrenActions

These are the child OUs and AWS accounts contained by this OU.

Organizational structure

Account created/joined date

☐

 Piyush-Prod

154335822557 | siriusdev19+1@gmail.com

Created 2026/03/01

uat

Delete

Rename

An organization unit (OU) is a container for accounts within a root. An OU can contain other OUs up to five levels deep. This enables you to create an inverted tree hierarchy. The structure has a root at the top and branches of OUs that reach down. The branches end in accounts that act as the leaves of the tree. [Learn more](#)

Organizational unit details

Name
uat

ID
ou-6yon-aymhbki5

ARN
arn:aws:organizations::596059883112:ou/o-jkuqsqduco/ou-6yon-aymhbki5

Children | Tags | Policies

Children

Actions

These are the child OUs and AWS accounts contained by this OU.

Organizational structure	Account created/joined date
--------------------------	-----------------------------

This resource is empty

✔ Successfully moved the AWS account 'Piyush' to the organizational unit 'uat'.
✕

uat

Delete
Rename

An organization unit (OU) is a container for accounts within a root. An OU can contain other OUs up to five levels deep. This enables you to create an inverted tree hierarchy. The structure has a root at the top and branches of OUs that reach down. The branches end in accounts that act as the leaves of the tree. [Learn more](#)

Organizational unit details

Name
uat

ID
ou-6yon-aymhbkiis

ARN
arn:aws:organizations::596059883112:ou/o-jkuqsqudeo/ou-6yon-aymhbkiis

Children
Tags
Policies

Children

Actions ▼

These are the child OUs and AWS accounts contained by this OU.

Organizational structure	Account created/joined date
☐ Piyush 097021215071 psblack+1@gmail.com	Created 2026/02/26

6. Illustrate the difference between permissions attached to the AWS applications; attached to the account within the organization and account outside the organization.

Ans:

IAM Policies : These permissions are managed using AWS Identity and Access Management (IAM). IAM policies define what actions a user, group, or role can perform on AWS resources.

These permissions are typically attached to:

- **IAM Users**
- **IAM Groups**
- **IAM Roles**
- **AWS services (applications)** such as EC2, Lambda, or ECS.

For example:

- An **EC2 instance** may have an **IAM role** attached that allows it to read objects from an S3 bucket.
- A **Lambda function** may have permission to write logs to CloudWatch.

Account Level Permissions (SCP) :

When multiple AWS accounts are managed using **AWS Organizations**, administrators can apply Service Control Policies (SCPs). SCPs define the maximum allowed permissions for accounts inside the organization.

Important characteristics:

- SCPs do not grant permissions.
- They limit or restrict permissions that IAM policies can grant.
- They apply to entire AWS accounts or Organizational Units (OUs).

Permissions Outside Organisation:

When an AWS account needs to access resources in another account that is not part of the same organization, AWS uses cross-account IAM roles.

In this model:

1. A role is created in the target account.
2. A trust policy allows the external account to assume that role.
3. The external account uses AWS STS (Security Token Service) to obtain temporary credentials.

7. What happens when you invite an account to an AWS Organization? Is any IAM role created automatically? Why is a role created for newly created accounts within the organization, and what is its relevance?

Ans:

Invite Case: No IAM role is automatically created when you invite an existing AWS Account to join your organisation. SCP's begin applying to the account as soon as it becomes a member. We have to manually create an IAM role in that account and set up the trust policy if you want cross-account access.

Add Account: AWS automatically creates a cross-account role (by default named OrganizationAccountAccessRole) when you create a new account within AWS Organisations straight from Management Account. This role has Administrator Access attached and trusts the management account. It aims to use sts:AssumeRole to grant the main account instant administrative access to the new member account without requiring credentials to be shared.

8. In a Service Control Policy (SCP), can we define multiple values for a single key (for example, key:value pairs)? Is it possible to enforce allowed values directly within an SCP policy?

Ans:

Yes, in a Service Control Policy (SCP) you can define multiple values for a single condition

key . AWS allows a key to contain list of values in the policy condition. However SCPs cannot directly enforce "allowed values". SCPs are guardrails that primarily deny actions , not grant permissions. Therefore to enforce allowed values we usually :

1. Deny everything except specific values
2. Use Allow + explicit Deny conditions.

DenyRegion

DeleteEdit policy

Policy details

Name
DenyRegion

ARN
arn:aws:organizations::596059883112:policy/o-jkuqsqduao/service_control_policy/p-74js3pzv

Policy type
Service control policy (customer managed)

Description
-

ContentTargetsTags

Content

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "DenyNVirginiaRegion",
      "Effect": "Deny",
      "Action": "*",
      "Resource": "*",
      "Condition": {
        "StringNotEquals": {
          "aws:RequestedRegion": [
            "us-east-1",
            "ap-south-1"
          ]
        }
      }
    }
  ]
}
```

Children | Tags | **Policies**

You have enabled the following 2 policy types out of the [13 available](#) to the organization.

Resource control policies

Resource control policies (RCPs) enable central administration over the permissions that determine the baseline level of access that all identities have to any resources you share across the accounts in your organization. Policies attached to the root or to OUs can be inherited by child OUs and accounts. [Learn more](#)

▼ Applied policies (2)

Detach

Attach

	Name	Source	Description
☐	RCPSFullAWSAccess (AWS managed policy)	Attached directly	Allows access to every resource
☒	RCPSFullAWSAccess (AWS managed policy)	Attached to Root	Allows access to every resource

Service control policies

Service control policies (SCPs) enable central administration of the permissions available within the accounts in your organization. [Learn more](#)

▼ Applied policies (3)

Detach

Attach

	Name	Source	Description
☐	DenyRegion	Attached directly	-
☐	FullAWSAccess (AWS managed policy)	Attached directly	Allows access to every operation
☒	FullAWSAccess (AWS managed policy)	Attached to Root	Allows access to every operation

Resource created successfully in us-east-1 allowed region.

The screenshot shows the AWS Management Console interface for the 'Piyush-Prod' account in the 'United States (N. Virginia)' region. The 'Instances' page displays a single instance with the following details:

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
Allowed region	i-0444b971f134d74df	Running	t2.micro	Initializing	View alarms	us-east-1c	ec2-100-54-227-32.co...	100.54.227.32	-

Resource creation denied in eu-north-1 region.

The screenshot shows the AWS Management Console interface for the 'Piyush-Prod' account in the 'Europe (Stockholm)' region. The 'Instances' page displays an error message:

You are not authorized to perform this operation. User: arn:aws:sts::154335822557:assumed-role/OrganizationAccountAccessRole/piyush-admin is not authorized to perform: ec2:DescribeInstances with an explicit deny in a service control policy: arn:aws:organizations::596059883112:policy/o-jkuqsduco/service_control_policy/p-74js3pzw

9.What are Amazon S3–related policies within AWS Organization policies? How do they differ from bucket policies and IAM policies?

Ans:

Within AWS Organizations, there are S3-related policies called Resource Control Policies (RCPs) (available in newer AWS Organizations features). These policies allow administrators to control how Amazon S3 resources are used across accounts in an organization.

They are used to enforce organization-wide security rules for S3, such as:

- Enforcing encryption for S3 objects
- Restricting public access
- Controlling who can create or access buckets

These policies apply centrally across multiple AWS accounts, unlike IAM or bucket policies which operate at the individual account or resource level.

In contrast, IAM policies are identity-based policies attached to users, groups, or roles to define allowed actions, while bucket policies are resource-based policies attached directly to an S3 bucket to control who can access that specific bucket. Thus, organization policies operate at the highest governance level, while IAM and bucket policies operate at account and resource levels respectively.

10. When creating an Amazon RDS instance with a specific storage type (e.g., gp2), which API action is evaluated in the background? Is it ec2:CreateVolume or an RDS-specific API? How can we restrict RDS StorageType creation ?

Ans:

When creating an Amazon RDS instance with a specific storage type such as gp2, AWS evaluates the RDS-specific API action **rds:CreateDBInstance**. Although RDS uses Amazon EBS volumes internally for storage, the user does not directly interact with EC2 APIs like **ec2:CreateVolume**. The provisioning of underlying storage is managed by the RDS service itself, so IAM and SCP policies must allow RDS actions rather than EC2 volume actions. We can restrict the RDS StorageType creation using IAM policies or Service Control Policies (SCPs) by applying conditions on storage type parameter used during database creation.

11. What are Service-Linked Roles? Which AWS services use service-linked roles to communicate between the management (organization) account and member (child) accounts?

Ans:

Service-Linked Roles (SLRs) are a special type of IAM role that is automatically created and managed by an AWS service. These roles allow an AWS service to perform actions in your account on your behalf.

They have:

- A predefined trust relationship with a specific AWS service
- Permissions defined by AWS
- They cannot be modified except for description or deletion

Service-linked roles are commonly used in **AWS Organizations** to allow services to manage resources across the management account and member accounts.

AWS Service	Service-Linked Role Created
AWS Organizations	AWSServiceRoleForOrganizations
Amazon EC2 Auto Scaling	AWSServiceRoleForAutoScaling
Amazon ECS	AWSServiceRoleForECS
Amazon EKS	AWSServiceRoleForAmazonEKS
AWS Config	AWSServiceRoleForConfig
AWS CloudTrail	AWSServiceRoleForCloudTrail
Amazon GuardDuty	AWSServiceRoleForAmazonGuardDuty
AWS Security Hub	AWSServiceRoleForSecurityHub
AWS Trusted Advisor	AWSServiceRoleForTrustedAdvisor
AWS Elastic Load Balancing	AWSServiceRoleForElasticLoadBalancing
Amazon Lex	AWSServiceRoleForLexBots
Amazon SageMaker	AWSServiceRoleForSageMaker
Amazon Macie	AWSServiceRoleForAmazonMacie
AWS Access Analyzer	AWSServiceRoleForAccessAnalyzer
AWS Systems Manager	AWSServiceRoleForAmazonSSM
AWS Backup	AWSServiceRoleForBackup
AWS Firewall Manager	AWSServiceRoleForFMS
AWS WAF	AWSServiceRoleForWAF

12. What is a Delegated Administrator in AWS Organizations? Which AWS services support delegated administrator functionality?

Ans:

A Delegated Administrator in AWS Organization is a member (child) account that is given permission to manage a specific AWS service across the entire organization.

Normally, only the Management Account can configure organization-wide services.

With Delegated Administrator, the management account can assign another account to administer a service for all accounts in the organization.

This helps with:

- Centralized management
- Security separation
- Least privilege for the management account

AWS Service	Purpose
AWS Config	Organization-wide configuration monitoring
AWS CloudFormation StackSets	Deploy stacks across accounts
Amazon GuardDuty	Centralized threat detection
AWS Security Hub	Aggregates security findings
Amazon Macie	Data security and classification
AWS Firewall Manager	Manage WAF and Shield across accounts
AWS IAM Access Analyzer	Analyze resource sharing across accounts
AWS Audit Manager	Organization-wide audit management
AWS Backup	Centralized backup management
AWS Detective	Security investigation across accounts
AWS Service Catalog	Manage shared service catalogs
AWS Systems Manager	Manage instances across accounts

13. What are AWS Systems Manager (SSM) Documents? What are their common use cases? Illustrate

Ans:

AWS Systems Manager (SSM) Documents are JSON or YAML formatted documents that define actions AWS Systems Manager performs on managed resources such as EC2 instances, on-premises servers, or other AWS resources.

They act as automation scripts or configuration instructions that Systems Manager executes. SSM documents are used with features like:

- Run Command
- Automation
- State Manager
- Session Manager
- Patch Manager

14. Explain the process of configuring cross-region and cross-account backups using AWS Backup. Also describe the steps involved in restoring resources from these backups.

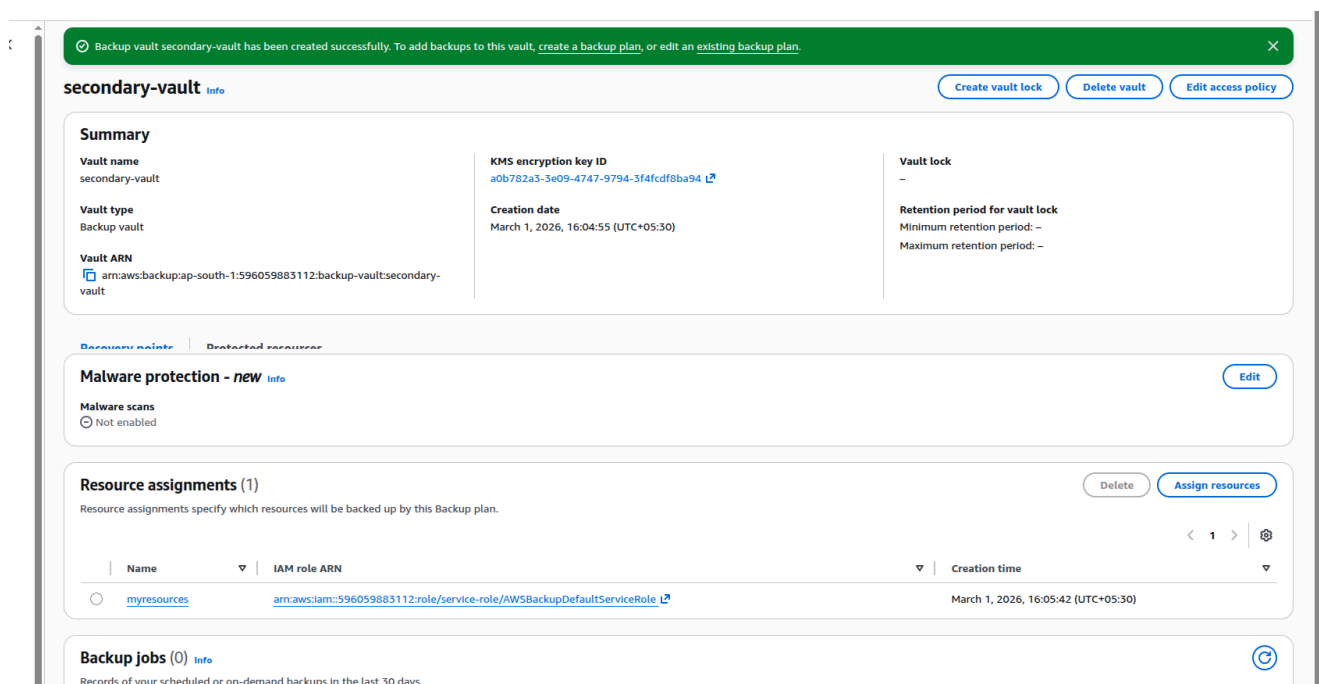
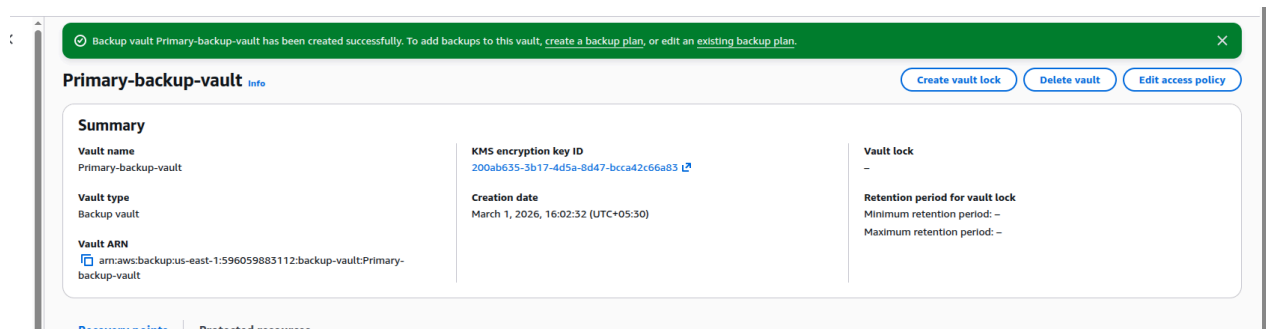
Ans:

AWS Backup allows centralized backup management across AWS services. It supports cross-region backup and cross-account backup to improve disaster recovery and data protection.

- **Cross-Region Backup:** Copies backups from one AWS region to another region.
- **Cross-Account Backup:** Copies backups from one AWS account to another account within the same AWS Organization.

This ensures that if a region fails or an account is compromised, backups still exist in another location.

To restore, AWS Backup retrieves the recovery point from the vault and recreates the source.



15. Understand SAML and Auth2 difference.

Ans:

SAML (Security Assertion Markup Language) is an XML-based authentication protocol used for enterprise Single Sign-On (SSO), allowing users to authenticate once and access multiple applications through an identity provider. OAuth 2.0 is an authorization framework that enables applications to access user resources from another service using access

tokens without exposing user credentials. While SAML focuses primarily on authentication and identity federation, OAuth 2.0 focuses on delegated authorization for API access.